The mixed complete-Pick interpolation question was answered negatively by arXiv:2210.14477

Literature-status packet for arXiv:1701.04885

11 August 2026

Status: literature already answered. Tsikalas explicitly restates the Aleman–Hartz–McCarthy–Richter question, constructs counterexamples to its proposed equivalence, and proves the correct replacement criterion. This is a retrieval result, not a new counterexample produced by the present run.

Original question

Aleman, Hartz, McCarthy, and Richter ask in the final paragraph of Section 6, "Applications and Questions," page 21 of the arXiv source PDF [1]:

> If s is a normalized complete Pick kernel and $\ell = gs$ for some positive-semidefinite kernel g, is a sequence $(\lambda_i) \subseteq X$ interpolating for $\mathrm{Mult}(\mathcal{H}_s, \mathcal{H}_\ell)$ if and only if it is a Carleson sequence for $\mathcal{H}_s$ and is $\mathcal{H}_\ell$-separated?

The source had proved this under additional hypotheses, notably when the target kernel is a power of a complete Pick kernel, but left the general statement open.

Decisive later answer

Georgios Tsikalas, *Interpolating sequences for pairs of spaces*, arXiv:2210.14477; J. Funct. Anal. 285 (2023), 110059 [2], explicitly labels the same statement as Question 1.1 and credits it to the four source authors. The abstract likewise says that the paper answers their question.

The answer is **no in general**. Supporting Theorem 1.5 constructs a kernel ℓ with a normalized complete Pick factor s such that, for every $n \geq 2$, there is a sequence satisfying the $\mathcal{H}_s$ Carleson condition and n-weak separation by ℓ, but not $(n+1)$-weak separation and hence not $\mathrm{Mult}(\mathcal{H}_s, \mathcal{H}_\ell)$ interpolation. Taking $n = 2$ gives exactly a sequence that is pairwise (ordinary weakly) separated and $\mathcal{H}_s$-Carleson but is not interpolating, disproving the proposed equivalence.

Supporting Theorem 1.4 gives the corrected full characterization:

$$(\lambda_i) \text{ is } \mathrm{Mult}(\mathcal{H}_s, \mathcal{H}_\ell)\text{-interpolating} \quad \Longleftrightarrow \quad \begin{cases} (\lambda_i) \text{ is } \mathcal{H}_s\text{-Carleson}, \\ (\lambda_i) \text{ is } n\text{-weakly separated by } \ell \text{ for every } n \geq 2. \end{cases}$$

Here n-weak separation uniformly bounds the distance of any one normalized ℓ-kernel from the span of any other $n-1$ kernels. Ordinary separation is only the case $n = 2$.

Provenance and scope

This is an unambiguous `literature_already_answered` match: the later author explicitly cites and restates the original question, gives a negative counterexample theorem, and replaces the false criterion by an exact one. The later paper also identifies important classes where ordinary separation does suffice, via its automatic-separation property. Those positive subclasses do not change the negative answer to the source's unrestricted question.

The packet contains the original paper as `source_paper.pdf` and the decisive later paper as `supporting_paper_2210.14477.pdf`. A matching run-ledger record accompanies this status packet.

References

[1] A. Aleman, M. Hartz, J. E. McCarthy, and S. Richter, *Interpolating sequences in spaces with the complete Pick property*, Int. Math. Res. Not. IMRN 2019, no. 12, 3832–3854; arXiv:1701.04885.

[2] G. Tsikalas, *Interpolating sequences for pairs of spaces*, J. Funct. Anal. 285 (2023), 110059; doi:10.1016/j.jfa.2023.110059; arXiv:2210.14477.